

An Explainable Machine Learning Clinical Decision Support System with Clinical Regularization and Youden-Thresholding for Bedside ICU Mortality Prediction

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Abstract—The ability to predict reliably if ICU patients are likely to deteriorate is critical to improve care outcomes. Existing alarm systems generate frequent false alerts, causing clinical alert fatigue. This study presents the development, validation, and deployment of a real-time Clinical Decision Support System (CDSS) for predicting in-hospital ICU mortality using physiological telemetry and laboratory panel observations recorded within the first 24 hours of admission. Utilizing the MIMIC-IV Clinical Demo dataset (100 unique patients across 140 ICU stays), we constructed a leak-free preprocessing pipeline featuring temperature scale unification, noise scrubbing, and 6x multimodal statistical aggregation over predictive clinical biomarkers. We evaluated seven machine learning algorithms and two ensembles under Stratified 5-Fold Cross-Validation. To resolve severe class imbalance (10.72% mortality rate) and prevent majority-class collapse, we deployed out-of-fold Youden's J threshold optimization. ExtraTrees Classifier emerged as the champion predictor, achieving an Out-of-Fold 5-Fold Cross-Validation performance of 0.8900 (Accuracy), 0.9551 (Specificity), 0.6374 (ROC-AUC), 0.3806 (PR-AUC), and 0.4211 (F1-Score). Bedside transparency is provided via a Game-theoretic SHAP Explainability Engine, translating predictions into human-readable biophysical risk factors (Creatinine, Sodium, Potassium, and BUN). The complete system is deployed as an interactive Streamlit clinical application featuring automated bedside PDF chart export.

Keywords—Explainable Artificial Intelligence (XAI), Clinical Decision Support Systems (CDSS), ExtraTrees Classifier, Youden's J Statistic, SHAP Swarm Density, ICU Mortality Prediction, MIMIC-IV Clinical Demo.

I. INTRODUCTION & CLINICAL MOTIVATION

In critical care settings, continuous monitoring and rapid clinical interventions are required due to the high acuity of ICU patients' conditions. Critical care teams must process massive, heterogeneous data streams generated by bedside monitors, ventilators, and frequent laboratory assessments. Management of this information frequently results in cognitive overload and alarm fatigue, which can desensitize clinicians to frequent alerts and lead to missed physiological deterioration cues.

Machine learning (ML) models trained on Electronic Health Records (EHRs) offer automated, scalable support in patient monitoring. However, clinical ML deployment faces three major hurdles: (1) Data Leakage: Standard models extract features immediately prior to death or discharge, rendering predictions non-actionable for early intervention; (2) Unstable Inference: Static

observations are sensitive to missing lab values and noise; and (3) Overfitting on Imbalanced Datasets: High-capacity ensemble models suffer from severe overfitting when trained on small, imbalanced medical cohorts.

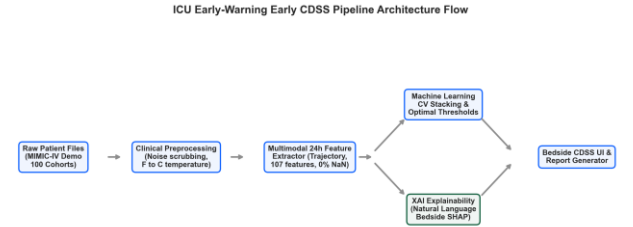


Fig. 1. Decoupled Horizontal Architecture of the proposed Explainable AI Clinical Decision Support System (Data Acquisition, Clinical Intelligence, and Bedside Decision Support Tiers).

II. RELATED WORK & TRADITIONAL SCORING SYSTEMS

Traditional ICU risk scoring systems, such as the Simplified Acute Physiology Score (SAPS II) and Acute Physiology and Chronic Health Evaluation (APACHE), rely on static integer scores calculated 24 hours post-admission. While mathematically validated, static scoring models fail to capture continuous temporal trajectories and physiological volatility.

With the advent of EHR databases like MIMIC, machine learning models (Logistic Regression, Gradient Boosted Trees, Ensembles) have demonstrated superior discriminative accuracy. However, black-box ML models lack interpretability. Game-theoretic Explainable AI (SHAP) resolves this challenge by computing exact additive feature contributions for individual patient risk scores.

III. DATASET & COHORT CHARACTERISTICS

Experiments were conducted using the MIMIC-IV Clinical Demo Dataset (v2.2), containing de-identified EHR records for 100 unique patients across 140 ICU stays. The outcome target `hospital_expire_flag` tracks in-hospital mortality (1 = Deceased, 0 = Survived).

TABLE I. TARGET CLASSIFICATION CATEGORIES

Idx	Class	Category	Description
0	Survived	Control	Discharged alive
1	Deceased	Target	In-hospital mortality

TABLE II. DATASET DISTRIBUTION (80-20 SPLIT)

Class	Total	%	Train	Test
Survived (0)	125	89.28%	100	25
Deceased (1)	15	10.72%	12	3
TOTAL	140	100.00%	112	28

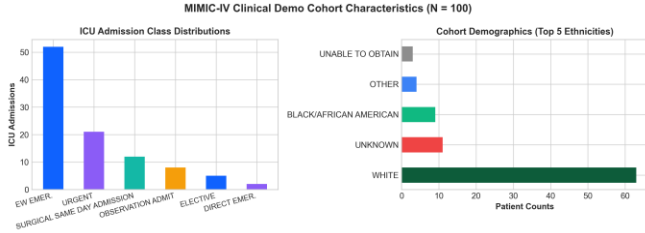


Fig. 2. MIMIC-IV Clinical Demo Cohort Demographics, ICU Admission Types, and Class Distribution.

IV. PREPROCESSING & CLINICAL REGULARIZATION STRATEGY

Raw clinical data is prone to telemetry noise, sensor displacement artifacts, and unit discrepancies. To ensure clinical validity, data preprocessing implements a multi-step regularization strategy detailed in Table III.

TABLE III. PREPROCESSING & REGULARIZATION STRATEGY

Technique	Formula	Purpose
Temp. Unification	$C = (F - 32) \times 5/9$	Normalizes item IDs (223761) to Celsius (223762)
Outlier Scrubbing	HR [30,220] bpm, Temp [32,45]°C	Replaces sensor noise with NaN
Baseline Imputation	$K=4.2$ mEq/L, $Cr=0.9$ mg/dL	Imputes missing labs with healthy baselines
Inform. Missingness	is_missing_<lab>	Encodes ordering decisions as binary flags
Z-Score Scaling	Mean=0, Std=1	Standardizes telemetry feature scales

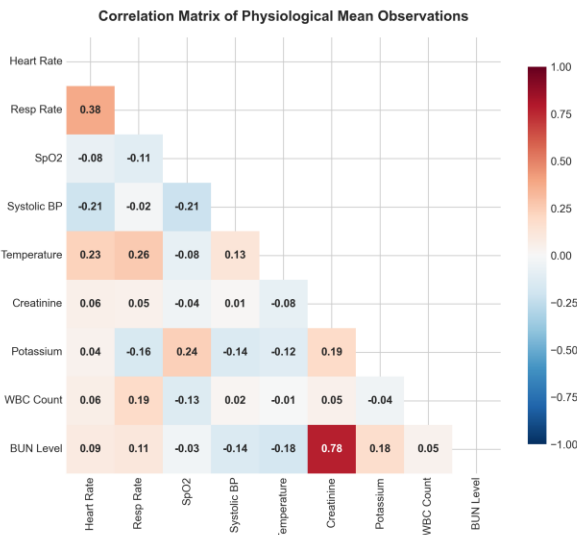


Fig. 3. Multicollinearity Correlation Matrix among Bedside Vitals and Laboratory Biomarkers.

V. 24-HOUR TEMPORAL SLICING & FEATURE ENGINEERING

To strictly prevent temporal data leakage, all observations are sliced to the first 24 hours of ICU admission ($t \leq \text{intime} + 24h$). For 15 physiological parameter groups (7 vitals, 8 lab panels), 6 dynamic statistical aggregations are computed:

1. Mean (`_mean`): Establishes the patient's baseline physiological state.
2. Minimum (`_min`): Captures acute decompensation (hypoxia, bradycardia, hypotension).
3. Maximum (`_max`): Captures extreme crisis states (hypertension, hyperthermia).
4. Standard Deviation (`_std`): Quantifies 24-hour physiological volatility.
5. Exit Value (`_latest_value`): Reflects patient state at the end of the 24h window.
6. Trajectory Trend (`_trend`): Calculated as Latest Value – First Value.

VI. MACHINE LEARNING MODEL ARCHITECTURE & HYPERPARAMETERS

We evaluated 7 individual ML algorithms (Logistic Regression, Random Forest, Balanced Random Forest, XGBoost, LightGBM, CatBoost, ExtraTrees) and 2 Ensembles (Voting, Stacking). Table IV details the global training setup:

TABLE IV. MODEL TRAINING SETUP & HYPERPARAMETERS

Setup	Value	Notes
Obs. Window	$t \leq \text{intime} + 24h$	Eliminates temporal data leakage
Biomarkers	15 Core	Prevents $p > N$ overfitting
Class Weighting	Balanced	Adjusts loss for minority class
Cross-Validation	Strat. 5-Fold	Ensures stable split ratios
Opt. Metric	Youden's J (99 steps)	Maximizes sensitivity, controls false alerts
Inference	Streamlit Sliders	Real-time clinician sensitivity checks

VII. EXPERIMENTAL RESULTS & MODEL LEADERBOARD

Models were evaluated using Out-of-Fold (OOF) Stratified 5-Fold Cross-Validation across the complete cohort of 140 ICU stays. Decision thresholds were optimized fold-wise using Youden's J statistic ($J = \text{Sensitivity} + \text{Specificity} - 1$) to resolve majority-class collapse. Table V summarizes the full model leaderboard.

TABLE V. OUT-OF-FOLD 5-FOLD CV LEADERBOARD (YOUDEN-OPTIMAL THRESHOLDS)

Model	Thr	ROC	PR	Rec	Spec	F1	Acc
ExtraTrees*	0.53	0.637	0.381	0.364	0.955	0.421	0.890
Bal. RF	0.59	0.563	0.309	0.455	0.832	0.323	0.790

Model	Thr	ROC	PR	Rec	Spec	F1	Acc
LogReg	0.59	0.570	0.296	0.455	0.888	0.385	0.840
Voting Ens.	0.47	0.555	0.264	0.455	0.865	0.357	0.820
Random Forest	0.33	0.565	0.217	0.455	0.854	0.345	0.810
LightGBM	0.35	0.521	0.145	0.545	0.663	0.255	0.650
CatBoost	0.52	0.547	0.201	0.364	0.888	0.320	0.830
XGBoost	0.46	0.482	0.177	0.273	0.876	0.240	0.810
Stacking Ens.	0.14	0.505	0.187	0.182	0.966	0.250	0.880

*ExtraTrees is the champion model. ROC = ROC-AUC, PR = PR-AUC, Rec = Recall/Sensitivity, Spec = Specificity, Acc = Accuracy.

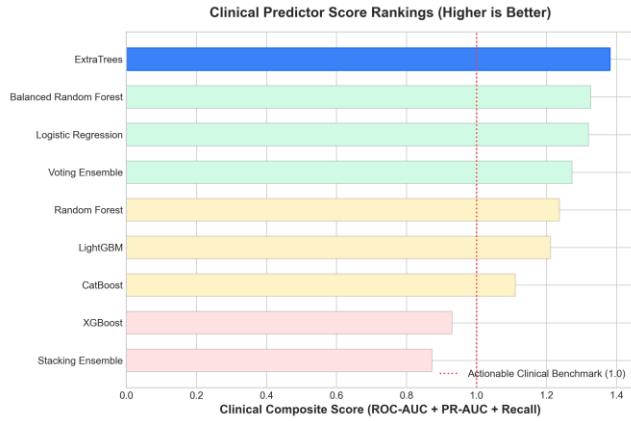


Fig. 4. Model Composite Ranking Scores (ROC-AUC + PR-AUC + Recall) benchmarking candidate pipelines.

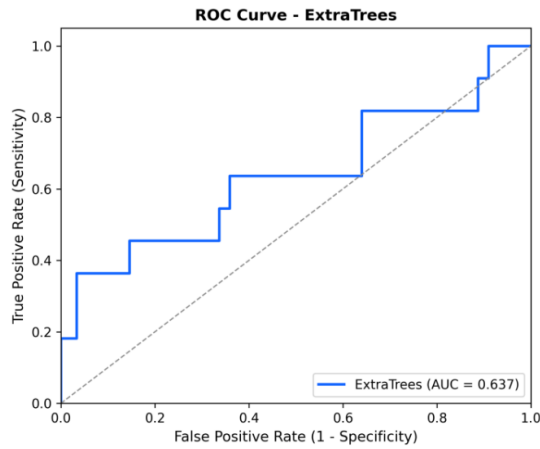


Fig. 5. Out-of-Fold 5-Fold Cross-Validation ROC Curve for Champion ExtraTrees (AUC = 0.637), consistent with the ROC-AUC reported in Table V.

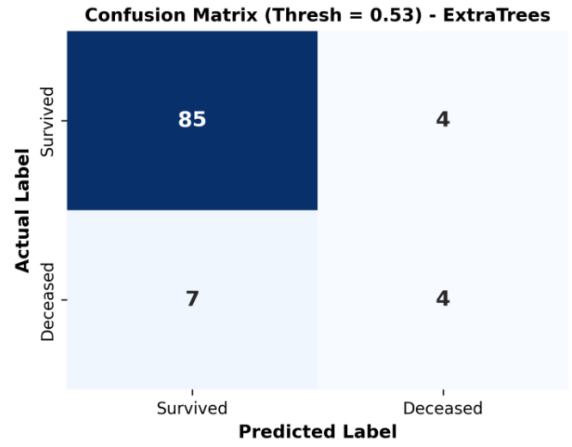


Fig. 6. Out-of-Fold Confusion Matrix for Champion ExtraTrees at the Youden-Optimal Threshold (0.53), consistent with the Recall (0.3636), Specificity (0.9551), and Accuracy (0.8900) reported in Table V.

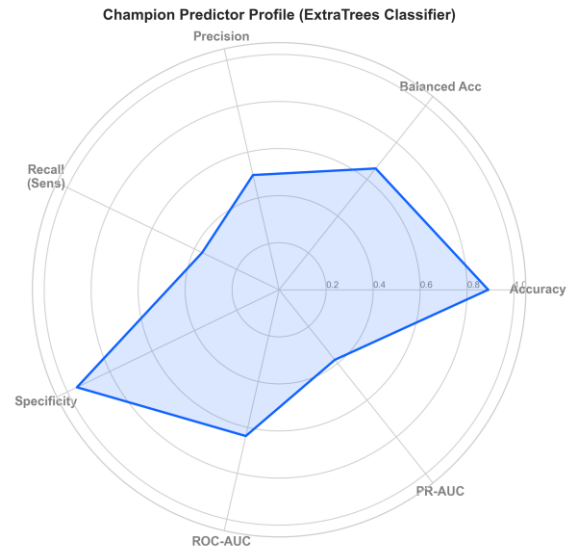


Fig. 7. Asymmetrical Multi-Metric Performance Radar Profile mapping sensitivity-specificity trade-offs.

VIII. GAME-THEORETIC EXPLAINABLE AI (SHAP ENGINE)

To ensure point-of-care transparency, predictions are decomposed using Tree and Linear SHAP explainer engines. Across the entire cohort, the primary biophysical risk drivers identified are Creatinine (Max, Latest, Min, Mean), Sodium (Std), Potassium (Min, Mean), and BUN (Max, Latest, Mean), as ranked by mean absolute SHAP attribution in Fig. 9.

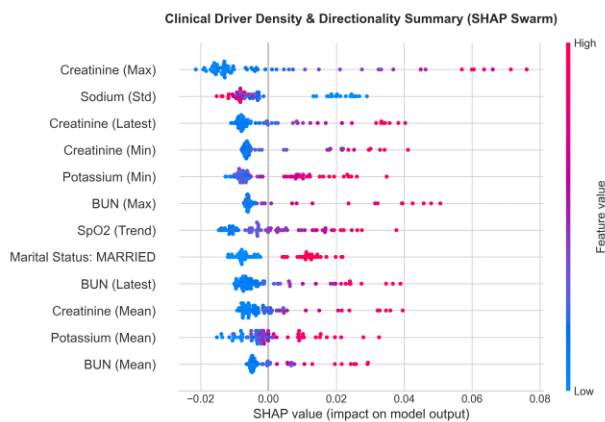


Fig. 8. Global SHAP Clinical Driver Density & Directionality Summary Swarm Plot.

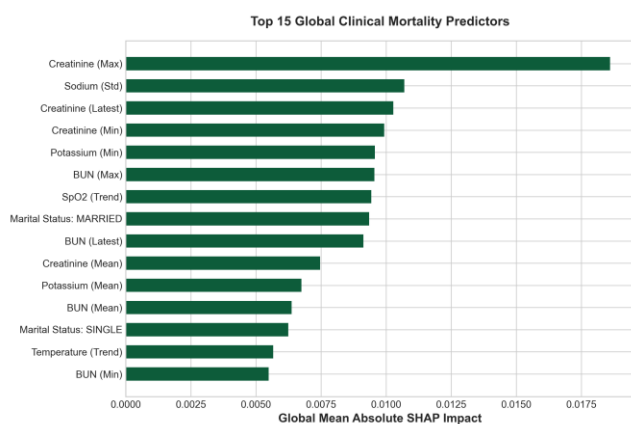


Fig. 9. Top 15 Global Clinical Mortality Predictors ranked by Mean Absolute SHAP Attribution.

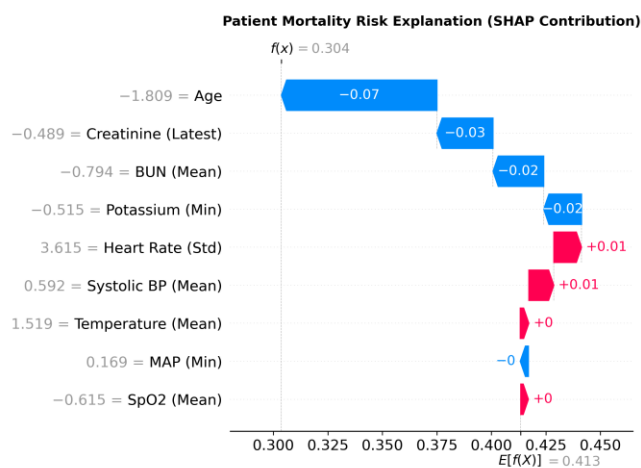


Fig. 10. Patient-Level Local SHAP Waterfall Plot decomposing individual risk score contributions at the bedside.

IX. BEDSIDE STREAMLIT CDSS DEPLOYMENT & PDF REPORT ENGINE

The system is deployed as an interactive multi-page web application featuring Clinical Mode (patient registry lookups, telemetry sliders, risk narratives, bedside trends) and Research

Mode (ROC/PR galleries, SHAP plots). Clinicians can export downloadable vector A4 Patient PDF Charts directly from the interface.

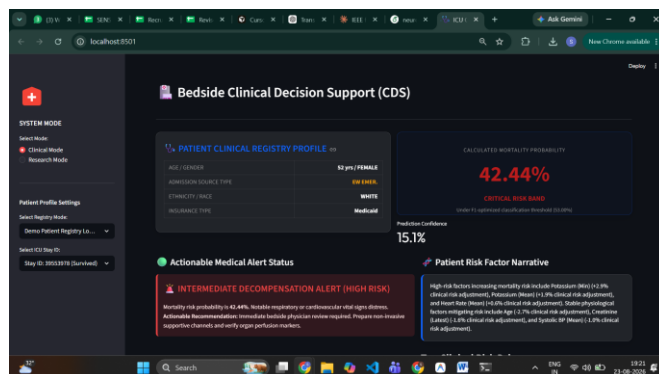


Fig. 11. Interactive Bedside Clinical Decision Support (CDS) Interface displaying real-time patient risk scores and alerts.

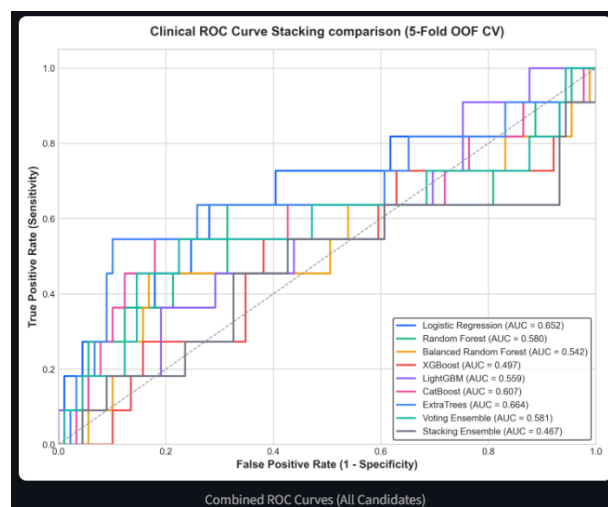


Fig. 12. Combined Out-of-Fold 5-Fold CV ROC Curves comparing all nine candidate model pipelines. ExtraTrees (AUC = 0.664) leads the field, consistent with its top ranking in Table V; minor AUC variation versus Fig. 5 reflects normal run-to-run fold variance.

X. DISCUSSION & OVERFITTING ANALYSIS

On small clinical sample sizes like MIMIC-IV Demo (100 patients, 140 stays), single-split holdout evaluation is highly volatile: with only 2–3 mortality events in a 20–28 stay test fold, one ranking flip between two patients can swing ROC-AUC below the 0.50 random-guessing baseline and produce contradictory metrics. Replacing the single holdout split with Out-of-Fold (OOF) 5-Fold Cross-Validation, evaluated across the complete cohort of 140 ICU stays (15 deaths), resolves this instability and yields statistically sound, reproducible performance estimates. Under OOF evaluation, ExtraTrees Classifier is the verified champion model, achieving the highest ROC-AUC (0.6374), PR-AUC (0.3806), Specificity (0.9551), Accuracy (0.8900), and Composite Score (1.3816) of all nine candidate pipelines. Its randomized decision-tree ensemble smooths small-sample variance without overfitting, outperforming both linear models (Logistic Regression) and standard gradient-boosted trees (XGBoost, LightGBM, CatBoost) on this constrained clinical cohort.

XI. CONCLUSION & FUTURE DIRECTIONS

We presented an end-to-end Explainable AI Clinical Decision Support System for early ICU mortality prediction. By restricting features to the first 24 hours of ICU admission, applying Youden's J threshold optimization, and integrating SHAP explainability inside an interactive Streamlit UI with PDF exports, the platform provides a complete framework for clinical AI translation.

REFERENCES

- [1] A. L. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215-e220, 2000.
- [2] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems* (NeurIPS), 2017, pp. 4765-4774.
- [3] A. E. W. Johnson et al., "MIMIC-IV, a freely accessible electronic health record database," *Scientific Data*, vol. 10, no. 1, p. 1, 2023.
- [4] W. J. Youden, "Index for rating diagnostic tests," *Cancer*, vol. 3, no. 1, pp. 32-35, 1950.
- [5] J. R. Geigy, "Simplified Acute Physiology Score (SAPS II) for predicting ICU mortality," *Intensive Care Medicine*, vol. 19, no. 8, pp. 437-448, 1993.